Technologies for Autonomous Vehicles
Mediapipe

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Outline

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- MediaPipe

Introduction

- EU Regulation (EU) 2019/2144 mandates that all new vehicles from 7th July 2024 are equipped with Driver Drowsiness and Attention Warning (DDAW) systems

- Euro NCAP developed the Driver Status Monitoring Spot Testing Guidance for assessing the capabilities of DDAW

- Car makers are using Euro NCAP document as guideline for DDAW development

Definitions

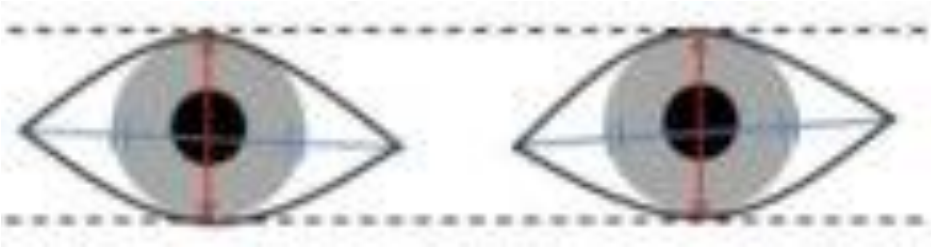
- Euro NCAP introduced few key concepts relevant for DDAW systems
- Impaired driving – A driver who is disconnected from the driving task or not in a physical state that is sufficient for safe driving, either due to distraction, fatigue or sickness
- Distraction – Anything that reduces the driver's focus on the primary task of driving/controlling the vehicle:
  - Long distraction – A single long duration distraction which takes the driver's gaze away from the forward road view
  - Short distraction / Visual Attention Time Sharing (VATS) – Repeated short duration gazes away from the forward road view, which cumulatively reduce the driver's awareness of the driving situation, until their attention returns to the driving task for long enough for them to fully assess the driving situation
  - Phone use – A subset of short distraction (VATS) where the object the driver's attention is shared with is their mobile phone

Definitions

- Fatigue – State of the driver where he/she is not awake enough to properly perform the driving task
- Drowsy – State of the driver where tiredness has an adverse effect of the driver's ability to focus on the driving task
- Microsleep – A microsleep is a temporary episode of sleep which may last up to several seconds
- Sleep – In this assessment sleep is considered as when a driver has been in a state of unconsciousness due to fatigue for a period of greater than a few seconds
- Unresponsive driver – Where a driver becomes unresponsive during driving, likely due to an onset of sudden sickness or extreme fatigue

Definitions

- Owl type movement – A shifting of visual attention away from the road and forward-facing position that is primarily achieved by head rotation followed by the eyes
- Lizard type movement – A movement in which the driver focuses on a task by moving primarily their eyeline away from the road with their head/face remaining in the forward-facing position
- Eye lid aperture – Distance between the point where the straight line drawn in the y-axis direction from the midpoint of line segment connecting the outer and inner corners of the driver's eye overlaps the lower edge of the upper eyelid and upper edge of the lower eyelid. Measured when driver is awake and attentive



Distraction

- Where applicable, Owl (head movement), Lizard (eye movement) and body lean looking behaviours are used to assess detection of driver distraction in three main areas: long distraction, short distraction (VATS) and phone use
- Long Distraction: A long distraction is considered a single long duration driver gaze away from the forward road to one consistent location of ≥ 3 seconds
- Short Distraction (VATS): A short distraction (or visual attention time sharing) event is considered to be repeated glances away from the forward road view either repeated towards one location, or to multiple different locations. A short distraction event is a build-up of multiple glances away from the forward road view and is considered to end when the driver's attention returns to the forward road view for a period long enough for the driver to fully interpret the road situation
- Phone Use: Phone use is considered to be a specific type of short distraction (or visual attention time sharing) event where the driver's repeated gaze is towards their mobile phone

Fatigue

- Fatigue is a typical behaviour that builds up over time
- Euro NCAP rewards detection of and response to different stages of fatigue: Drowsiness, Microsleep, and Sleep
- Drowsiness: State of the driver where tiredness has an adverse effect of the driver's ability to focus on the driving task
- Microsleep: A common symptom of a microsleep is an eye closure of < 3s although Euro NCAP recognises that non-eye closure microsleep events are possible
- Sleep: A driver is considered to be asleep in this assessment when displaying a continued eye closure >3 seconds

Unresponsive Driver

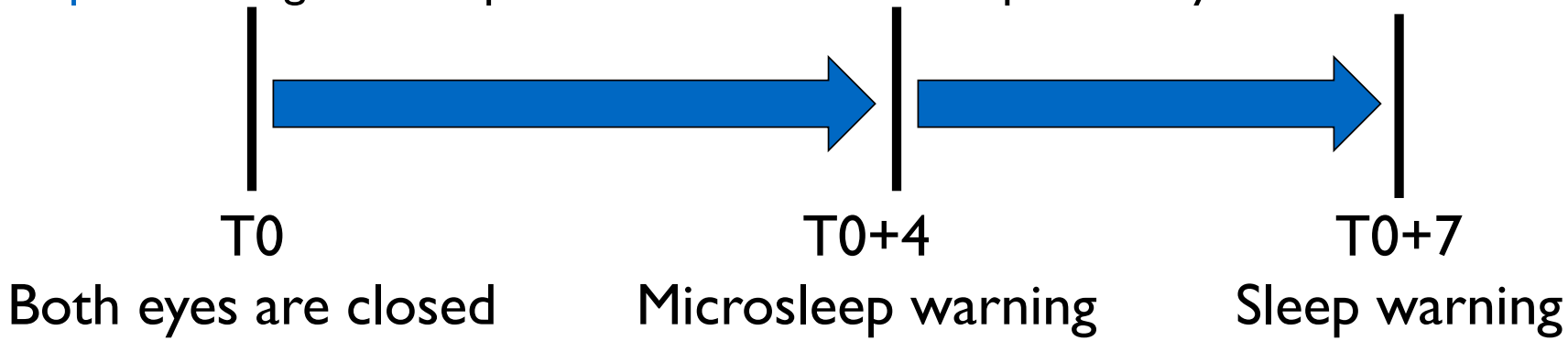
- Where a driver becomes unresponsive during driving, likely due to a sudden onset of sickness
- It is likely, but not certain, that initially an unresponsive driver will be determined as either distracted or asleep

Detection of distraction

- Owl long distraction – In case the driver maintains a gaze away from the forward road for at least 5 seconds, owl long distraction shall be reported to the driver until the driver return to gaze directly to the forward road
- Owl short distraction/own phone use – in case the driver head turns to gaze location and back to the forward road at a natural rate of movement for a cumulative 10 seconds within a 30 second time period, the owl short distraction shall be reported to the driver until the driver return to gaze directly to the forward road for a period of ≥ 2 seconds
- Lizard long distraction – same as owl, but the head remains in forward gaze position, while the eyes gaze at a different location
- Lizard short distraction/own phone use – same as owl, but the head remains in forward gaze position, while the eyes gaze at a different location

Detection of fatigue

- Microsleep – warning shall be provided if the driver keeps both eyes closed for at least 4 seconds
- Sleep – warning shall be provided if the driver keeps both eyes closed for at least 7 seconds



- Attention: Medical science tells us that sleep starts before eyes are closed, the mind can start wandering with the driver no longer paying attention to the road before closing the eyes
- Euro NCAP recognized this fact, but existing technology which is camera-based is not sufficient to match medical science with DMS needs (yet)

MediaPipe

- MediaPipe is a machine learning-based solution developed by Google
- It is a collection of pre-trained neural networks that can run on the users' devices, thanks to an API
- APIs allow us to perform various operations. One of the topics of interest for this lecture is the identification of facial landmarks

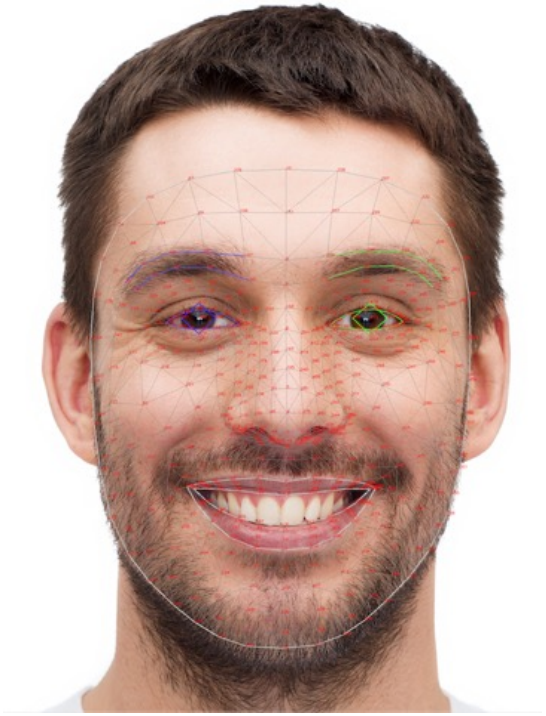
MediaPipe

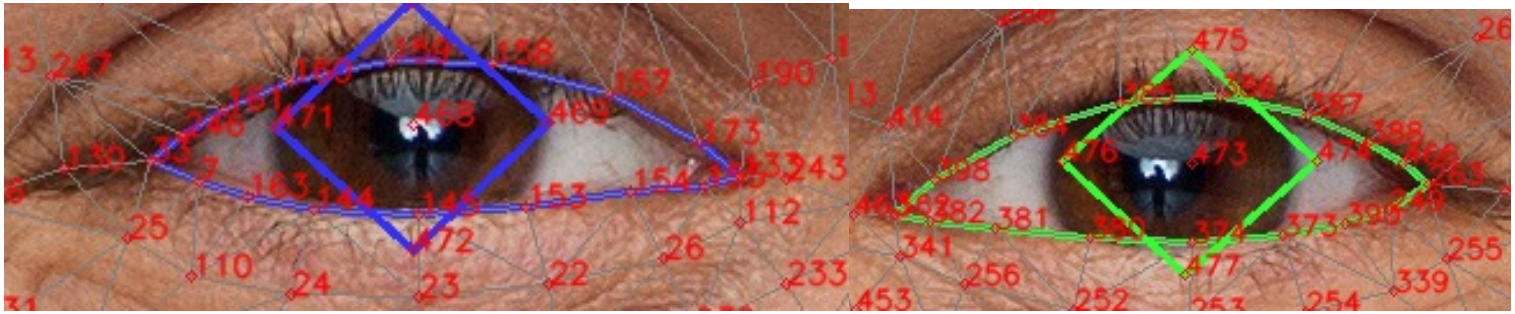
https://developers.google.com/mediapipe

Google MediaPipe FaceMesh

- It provides 3D coordinates of 468 facial landmarks position on an RGB camera image

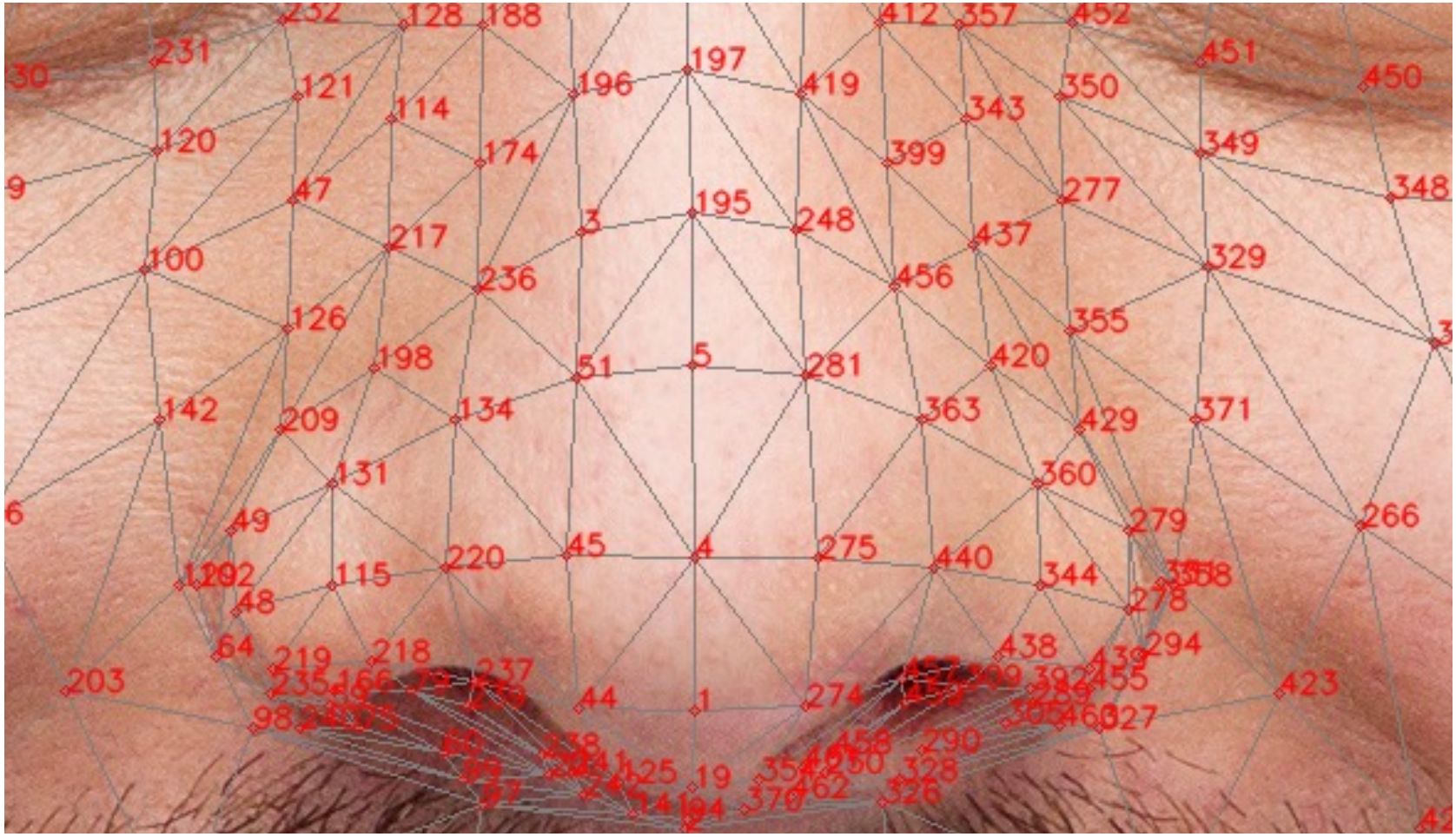
https://storage.googleapis.com/mediapipe-assets/documentation/mediapipe_face_landmark_fullsize.png



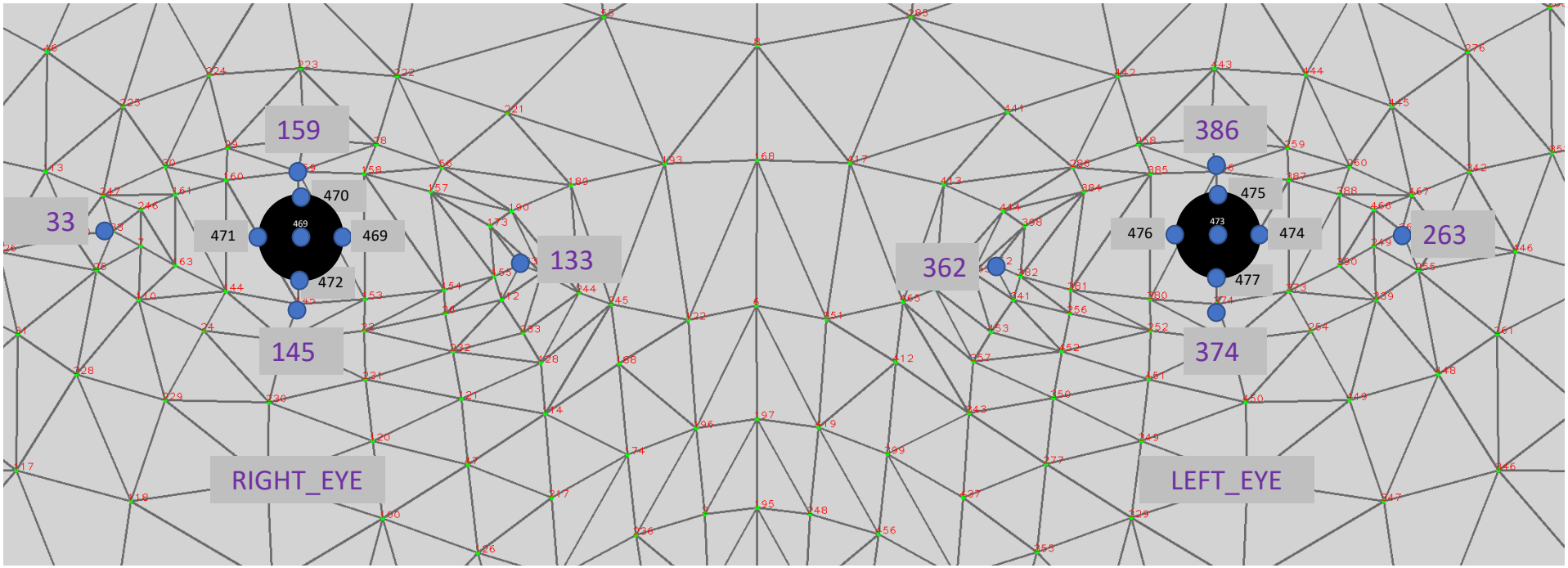


Each landmark is identified by a position index within the resulting structure (basically an array with 468 3D and 10 2D points).
The 468 3D points belong to the facial mesh, while the 10 2D ones belong to the iris tracking.

Google MediaPipe FaceMesh



Chosen landmarks



Points with an identifier <= 468 are from *FaceMesh*, while those >= 469 are from *Iris Tracking*

How does MediaPipe work?

- The whole frame from the camera is processed by a facial mesh detector, producing a reduced set of landmarks (like the eye center, ear regions, and nose tip). These landmarks are used to rotate a facial rectangle to align the line connecting the centers of the eyes with the horizontal axis of the rectangle
- The rectangle is cropped from the original image and resized to the facial mesh identification neural network (256x256 pixels for the full model and 128x128 for the smallest one).
- The mesh prediction neural network produces a vector of 3D landmark coordinates, which get mapped back into the original image coordinate system

A video input mode that optimizes performances is available, which allows a simplification of the face detection phase.

Reference: https://arxiv.org/abs/1907.06724

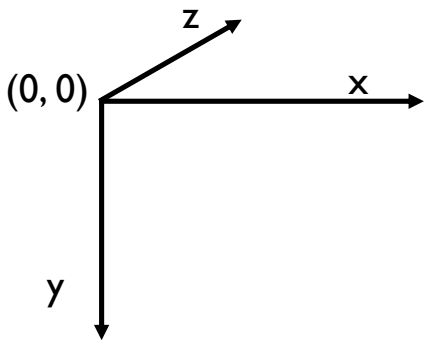
Coordinate system - 1

- *x* and *y* coordinates of the vertices correspond to the point locations in the 2D plane as given by the image pixel coordinates. The positions are normalized in [0; 1] range; hence, to obtain the position in pixels from the top-left corner, one must multiply *x* by the image width and *y* by the image height, then round the products to the nearest integer.

Reference: https://arxiv.org/abs/1907.06724

Coordinate system - II

- z coordinates are interpreted as the depth relative to a reference plane passing through the mesh's centre of mass. They are rescaled so that a fixed aspect ratio is maintained between the span of x-coordinates and z-coordinates, so that a face that is scaled to half its size has its depth range (nearest to farthest) scaled down by the same multiplier.



Reference: https://arxiv.org/abs/1907.06724

How has the model been trained?

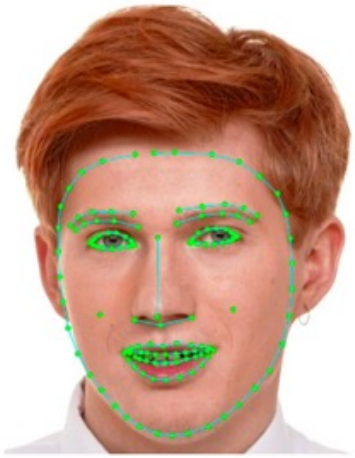
- The dataset is composed of 30K globally sourced in-the-wild camera photos taken from a variety of sensors in changing lighting conditions

- It has been further augmented :
  - With standard cropping and image processing primitives
  - By modeling camera sensor noise
  - By applying a randomized non-linear parametric transformation to the image intensity histogram (to help simulate marginal lighting conditions)

Reference: https://arxiv.org/abs/1907.06724

How has the model been trained?

- The ground truth consists of a 468 3D mesh point clouds for each image of the dataset
- Since creating the mesh by hand is too challenging, an iterative process is adopted, bootstrapped by a 3D model obtained by synthetic renderings and manual annotations of 2D landmarks corresponding to a small subset of the mesh vertices participating in a set of semantic contours, as shown in the figure below
- After the bootstrap, the model is trained for the first time
- From this first version, it was iteratively retrained on the facial meshes generated by its previous version.



Reference: https://arxiv.org/abs/1907.06724

Performance assesment

- MAD: Mean Absolute Distance between the predictions and the ground truth vertex location, normalized by the interocular distance (IOD)

- IOD: designed as the distance between the eye centers (estimated as the midpoints of the segment that connects the eye corners, to avoid gaze direction dependence)

- The normalization is aimed at avoiding factoring in the scale of the face

- Since z-coordinates are obtained exclusively from the synthetic supervision, the reported errors are only 2D. The IOD has been computed in 3D to account for possible yaw head rotations

Reference: https://arxiv.org/abs/1907.06724

Performance assesment

- The evaluation results are obtained on a set of 1700 images, while the speed estimations are based on the TensorFlow Lite GPU framework.

Model (input)	IOD MAD	Time, ms (iPhone XS)	Time, ms (Pixel 3)
Full (256×256)	3.96%	2.5	7.4
Light (128×128)	5.15%	1	3.4
Lightest (128×128)	5.29%	0.7	2.6

Table 1: Model performance characteristics

Reference: https://arxiv.org/abs/1907.06724

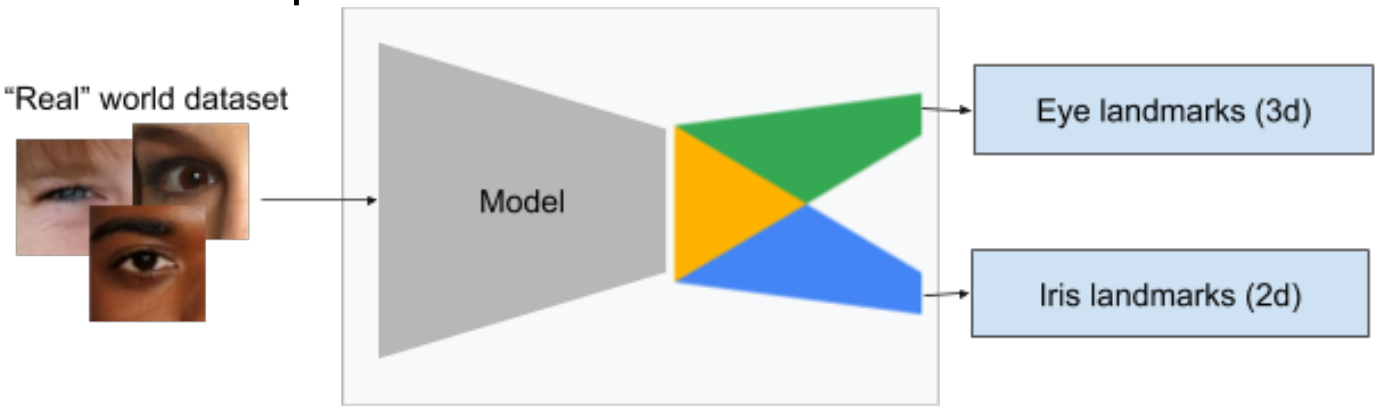
Iris tracking

Other than the 468 points mesh, MediaPipe can provide iris tracking through10 additional points.

Iris tracking is useful for:

- providing an estimation of where the driver is looking
- making it possible to determine the distance of the person depicted on the frame, allowing 3D positioning of the face in the space.

Annotated real world eye regions, with eyelid (red) and iris (blue) contours.



Training the model on the dataset

Reference: https://blog.research.google/2020/08/mediapipe-iris-real-time-iris-tracking.html

To run the script

- Load the libraries through the pip commands:

```
pip3 install opencv-python
pip3 install mediapipe
```

- Then run the script with:

```
python3 mediapipe_sample.py
```

MediaPipe settings

Option Name	Description	Value Range	Default Value
`running_mode`	Sets the running mode for the task. There are three modes: IMAGE: The mode for single image inputs. VIDEO: The mode for decoded frames of a video. LIVE_STREAM: The mode for a livestream of input data, such as from a camera. In this mode, resultListener must be called to set up a listener to receive results asynchronously.	{IMAGE, VIDEO, LIVE_STREAM}	IMAGE
`num_faces`	The maximum number of faces that can be detected by the the FaceLandmarker. Smoothing is only applied when num_faces is set to 1.	Integer > 0	1
`min_face_detection_confidence`	The minimum confidence score for the face detection to be considered successful.	Float [0.0,1.0]	0.5
`min_face_presence_confidence`	The minimum confidence score for the face presence in the face landmark detection.	Float [0.0,1.0]	0.5
`min_tracking_confidence`	The minimum confidence score for the face tracking to be considered successful.	Float [0.0,1.0]	0.5
`output_face_blendshapes`	Whether Face Landmarker outputs face blendshapes. Face blendshapes are used for rendering the 3D face model.	Boolean	False
`output_facial_transformation_matrixes`	Whether FaceLandmarker outputs the facial transformation matrix. FaceLandmarker uses the matrix to transform the face landmarks from a canonical face model to the detected face, so users can apply effects on the detected landmarks.	Boolean	False
`result_callback`	Sets the result listener to receive the landmarker results asynchronously when FaceLandmarker is in the live stream mode. Can only be used when running mode is set to LIVE_STREAM	ResultListener	N/A